SIMATS ENGINEERING

CSA17 – ARTIFICIAL INTELLIGENCE LAB

Day 1 – Question 10: 10. A* Algorithm

Aim: To find the shortest path from a start node to a goal node using A* search.

Python Program

```
import heapq

graph = {
    'A': [('B', 1), ('C', 4)],
    'B': [('C', 2), ('D', 5)],
    'C': [('D', 1)],
    'D': []
}
heuristic = {'A': 4, 'B': 3, 'C': 1, 'D': 0}

def astar(start, goal):
    pq = [(heuristic[start], 0, start, [start])]
    visited = set()
    while pq:
        f, cost, node, path = heapq.heappop(pq)
        if node == goal:
            print("Path:", path)
            print("Cost:", cost)
            return
        if node in visited:
            continue
        visited.add(node)
        for neighbour, weight in graph[node]:
            if neighbour not in visited:
                new_cost = cost + weight
                heapq.heappush(
                    pq, (new_cost + heuristic[neighbour],
                         new_cost, neighbour, path + [neighbour])
                )

astar('A', 'D')
```

Sample Output

```
Path: ['A', 'B', 'C', 'D']
Cost: 4
```

Result: The program demonstrates the stated Artificial Intelligence problem.